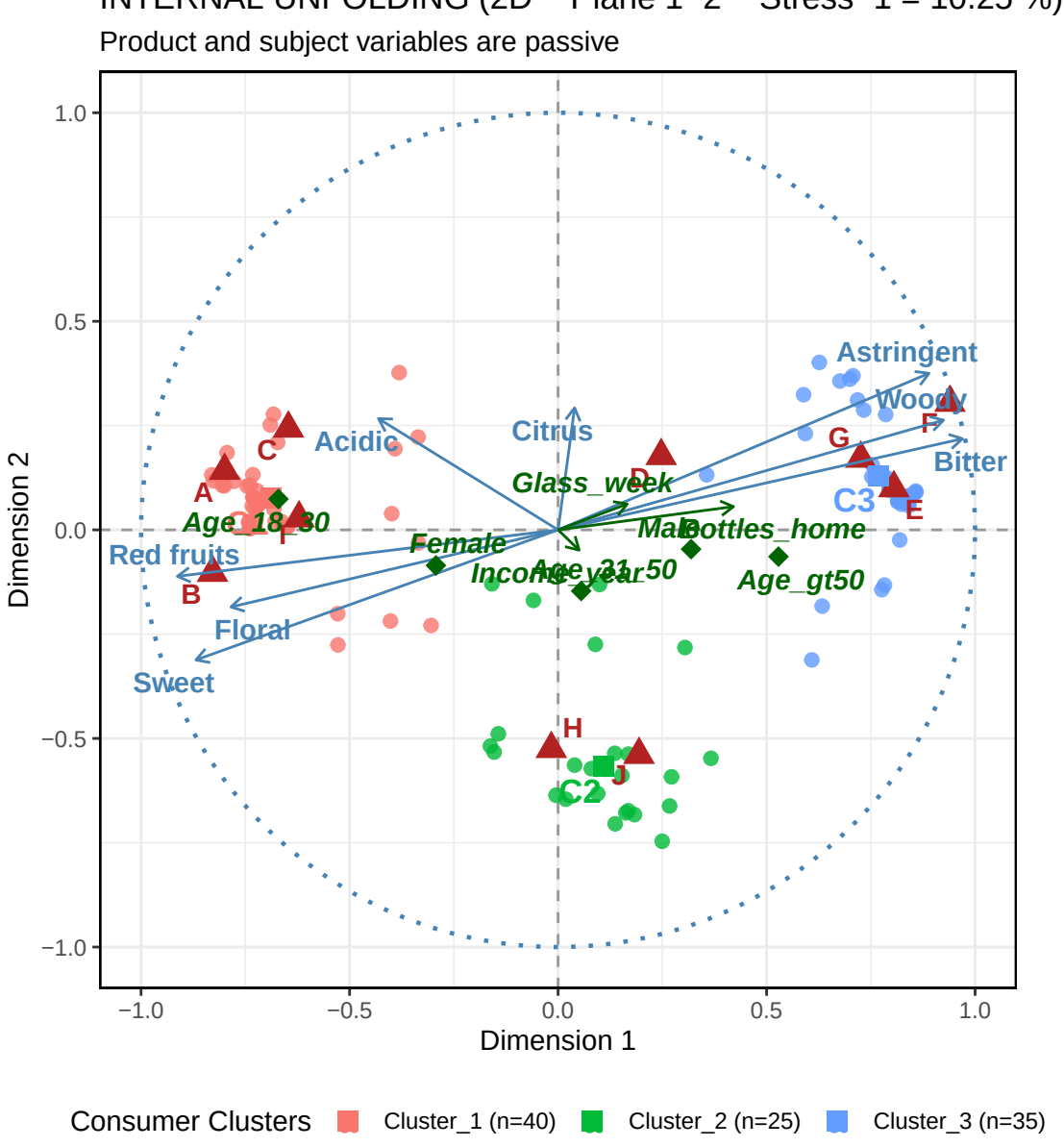
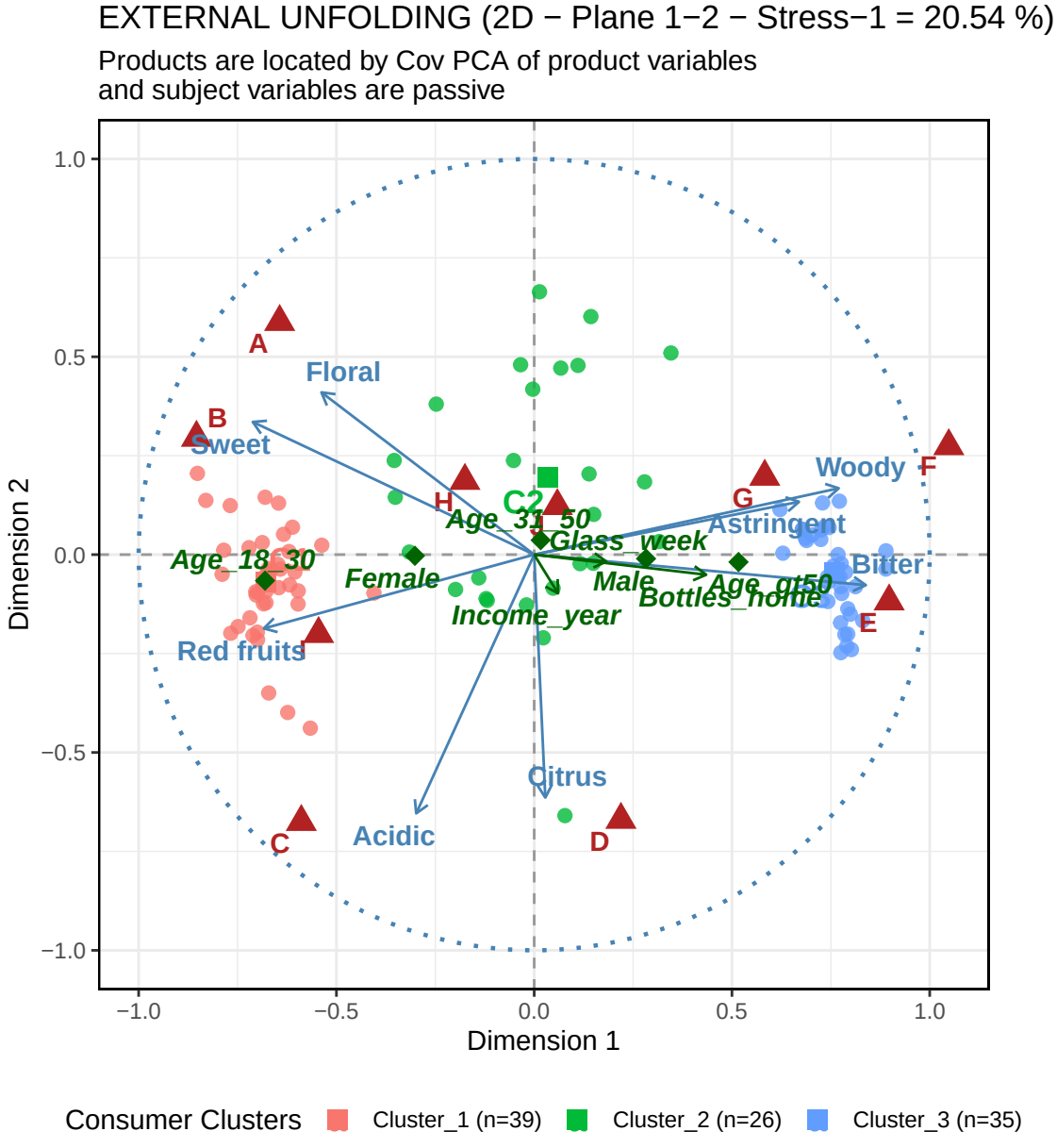
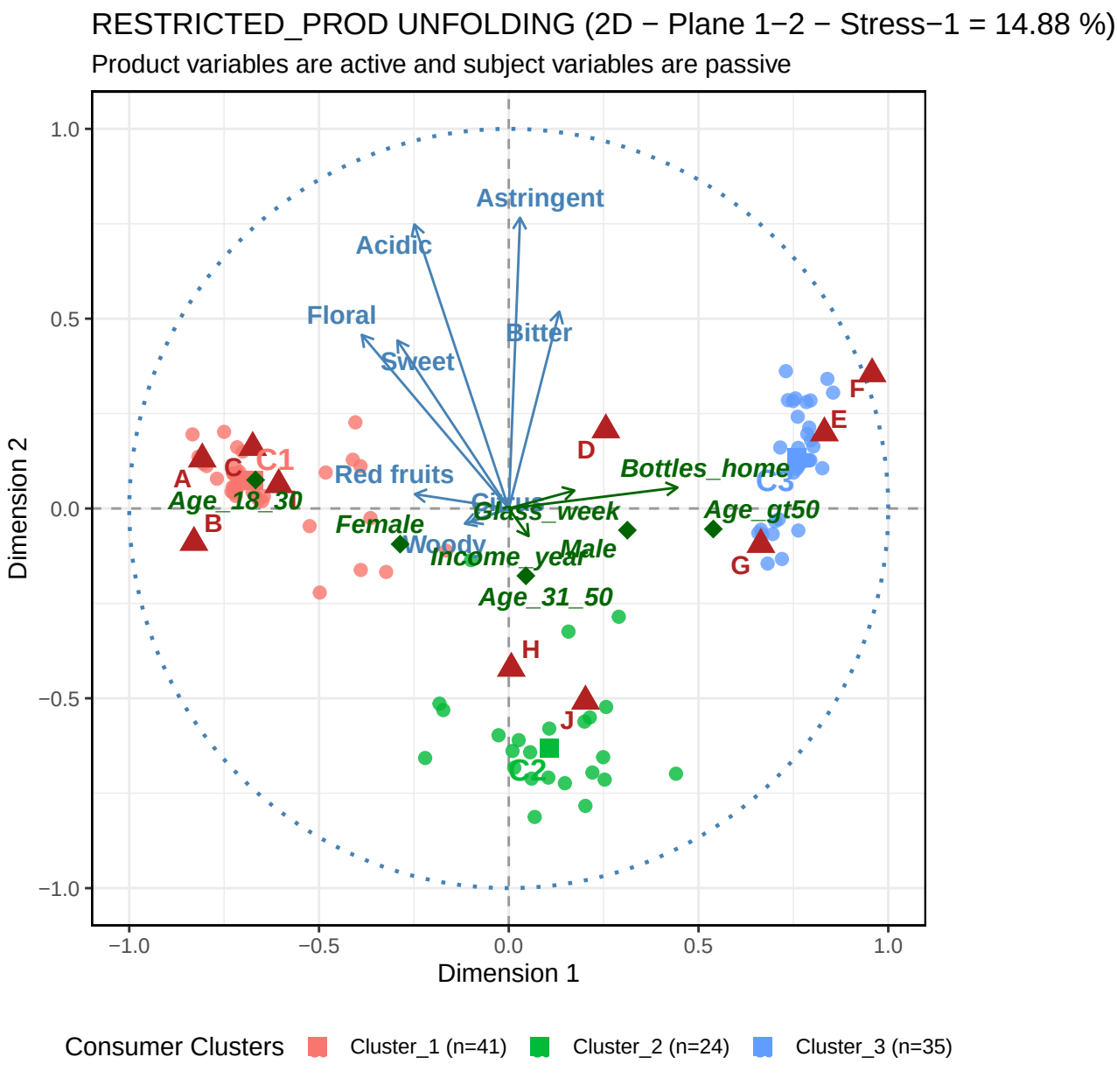
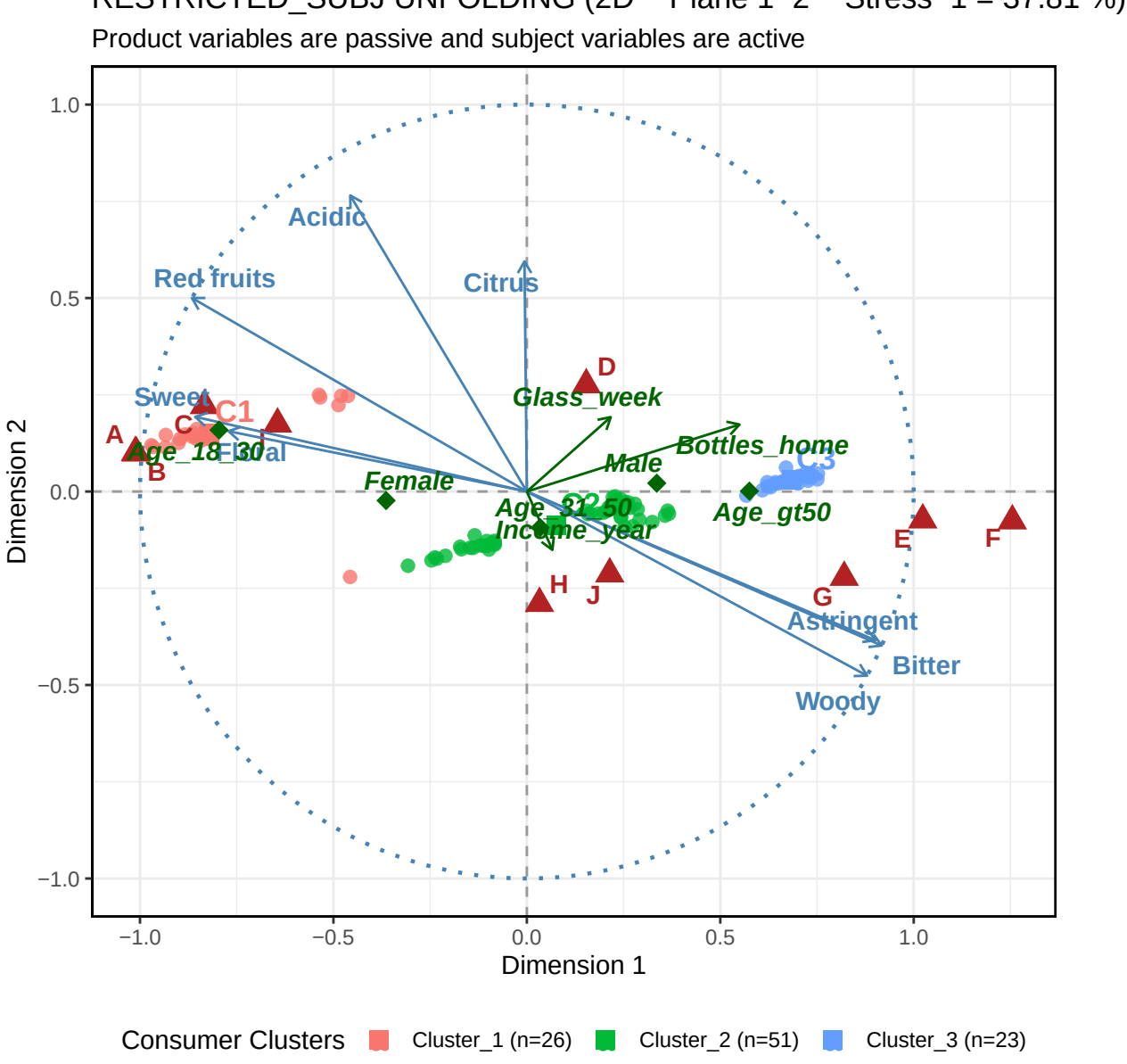
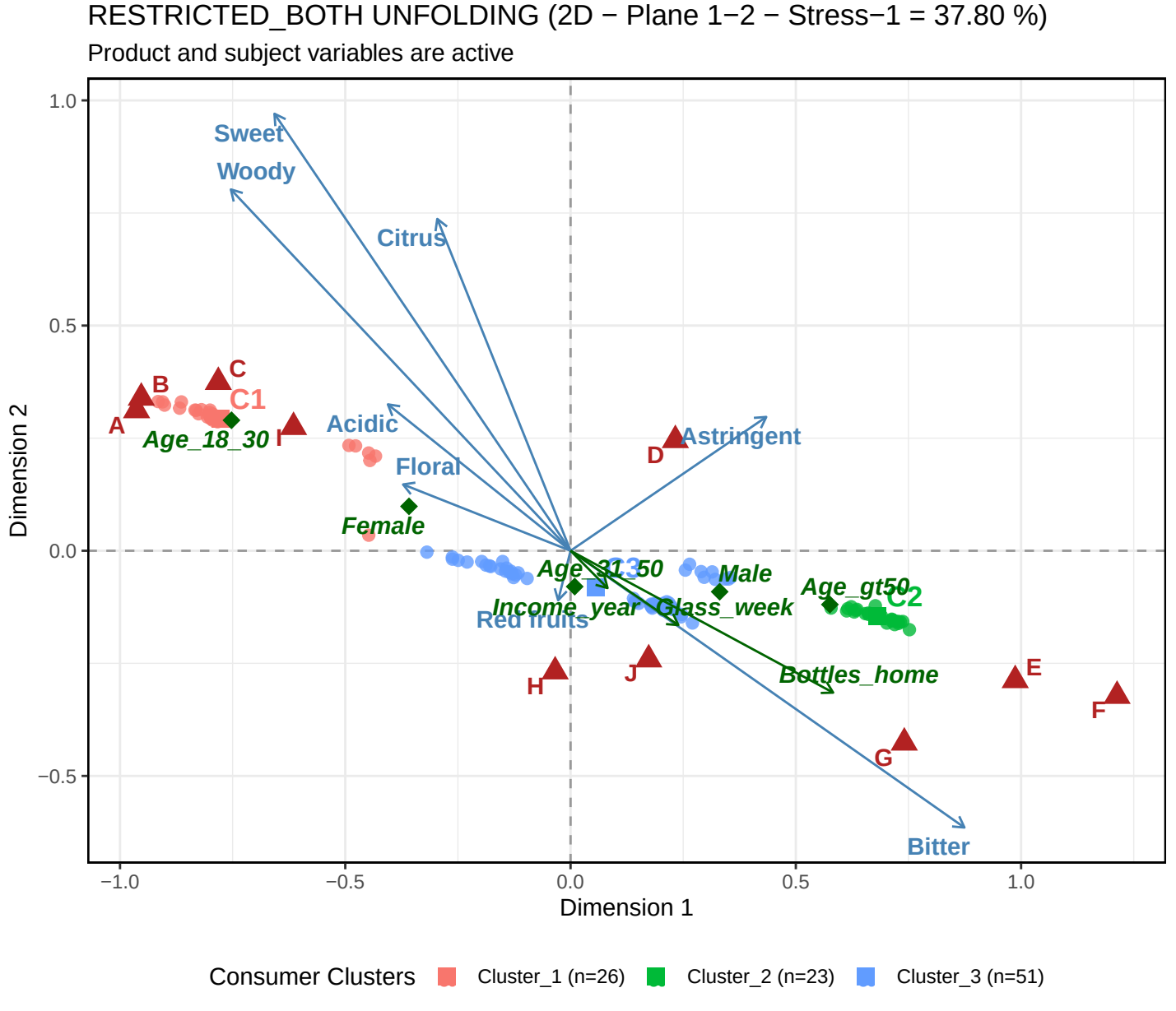

INTERNAL UNFOLDING (2D − Plane 1−2 − Stress−1 = 10.25 %)
Product and subject variables are passive
Dimension 2
Dimension 1
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Astringent
Woody
Bitter
Citrus
Acidic
Red fruits
Floral
Sweet
Glass_week
Female
Age_18_30
Income_year_21_50
Bottles_home
Age_gt50
A
B
C
D
E
F
G
H
J
C2
C3
Consumer Clusters
Cluster_1 (n=40)
Cluster_2 (n=25)
Cluster_3 (n=35)


EXTERNAL UNFOLDING (2D − Plane 1−2 − Stress−1 = 20.54 %)
Products are located by Cov PCA of product variables
and subject variables are passive
Dimension 2
Dimension 1
1.0
0.5
0.0
−0.5
−1.0
A
B
C
D
E
F
G
H
C2
Floral
Sweet
Red fruits
Acidic
Citrus
Woody
Astringent
Bitter
Age_18_30
Age_31_50
Age_gt50
Female
Male
Glass_week
Bottles_home
Income_year
Consumer Clusters
Cluster_1 (n=39)
Cluster_2 (n=26)
Cluster_3 (n=35)


RESTRICTED_PROD UNFOLDING (2D − Plane 1−2 − Stress−1 = 14.88 %)
Product variables are active and subject variables are passive
Astringent
Acidic
Floral
Sweet
Bitter
Red fruits
Citrus
Female
Woody
Class_week
Income_year
Male
Age_31_50
Bottles_home
Age_gt50
Age_18_30
A
B
C
C1
D
E
F
G
H
J
C2
C3
1.0
0.5
0.0
−0.5
−1.0
Dimension 2
Dimension 1
Consumer Clusters
Cluster_1 (n=41)
Cluster_2 (n=24)
Cluster_3 (n=35)


RESTRICTED_SUBJ UNFOLDING (2D − Plane 1−2 − Stress−1 = 37.81 %)
Product variables are passive and subject variables are active
Acidic
Red fruits
Citrus
Sweet
Glass_week
Bottles_home
Male
Female
Age_18_30
Age_31_50
Age_gt50
Income_year
Astringent
Bitter
Woody
A
B
C
C1
D
E
F
G
H
J
C3
Dimension 1
Dimension 2
1.0
0.5
0.0
−0.5
−1.0
Consumer Clusters
Cluster_1 (n=26)
Cluster_2 (n=51)
Cluster_3 (n=23)


RESTRICTED_BOTH UNFOLDING (2D − Plane 1−2 − Stress−1 = 37.80 %)
Product and subject variables are active
Sweet
Woody
Citrus
Acidic
Floral
Astringent
Bitter
Female
Age_18_30
Age_31_50
Age_gt50
Male
Income_year
Glass_week
Red fruits
Bottles_home
A
B
C
C1
C2
D
E
F
G
H
I
J
Dimension 1
Dimension 2
1.0
0.5
0.0
−0.5
−1.0
Consumer Clusters
Cluster_1 (n=26)
Cluster_2 (n=23)
Cluster_3 (n=51)